# DIGITAL IMAGE PROCESSING

## LECTURE # 12

## IMAGE COMPRESSION-I

6th May, 2026

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# Contact Information



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# Topics to Cover

- Fundamentals of Image Compression
- Background
- Overview
- Data VS Information
- Data Redundancy
- Coding Redundancy
- Interpixel Redundancy
- Psychovisual Redundancy
- Fidelty Criteria
- Image Compression Model

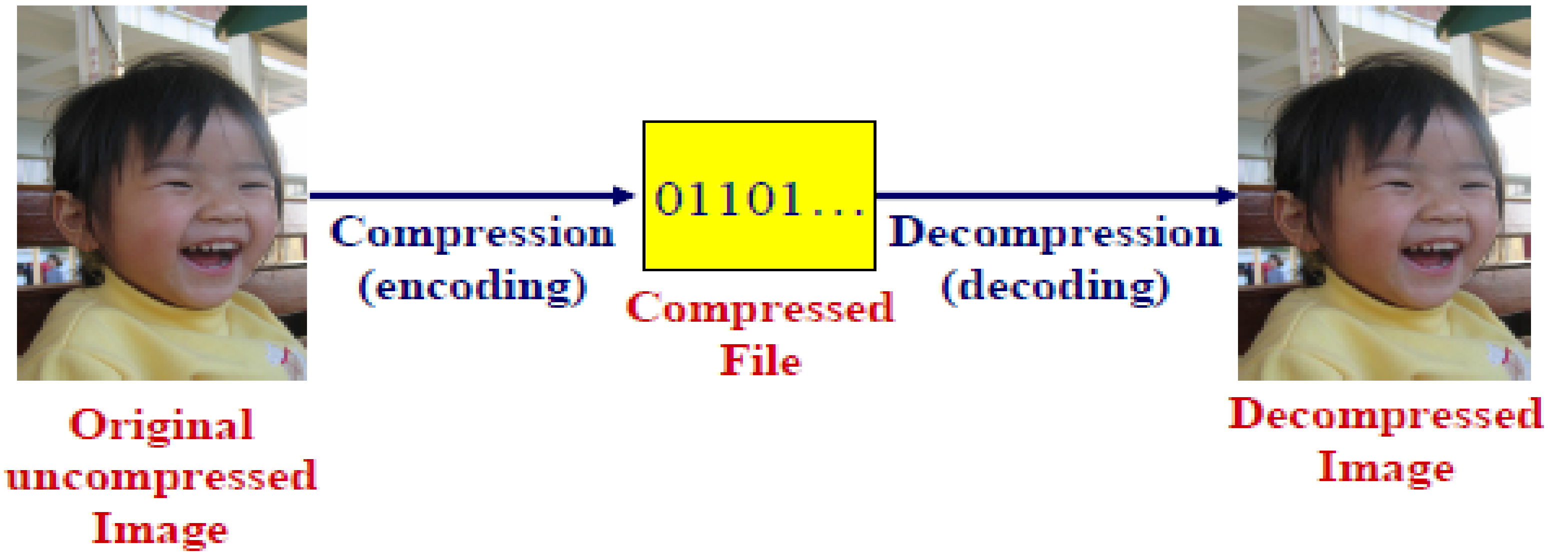
# Fundamentals of Image Compression



- Digital image contains a lot of data — far too much for practical applications (e.g., problem of storage and efficient transmission)

- Any data may be represented in different ways, image compression exploits redundancy to achieve reduction in the actual amount of data without (seriously) affecting the quality of the image

- **Image Compression-** reduce the size of image data file while retaining necessary information



# Background



- **Principal objective:**

  To minimize the number of bits required to represent an image.

- **Applications**

  - Transmission:

    Broadcast TV via satellite, teleconferencing etc.

  - Storage:

    Educational and business documents, medical images (CT, MRI and digital radiology), satellite images, weather maps, geological surveys, e.t.c)

# Overview



Image data compression methods fall into two common categories:

I. Information preserving compression

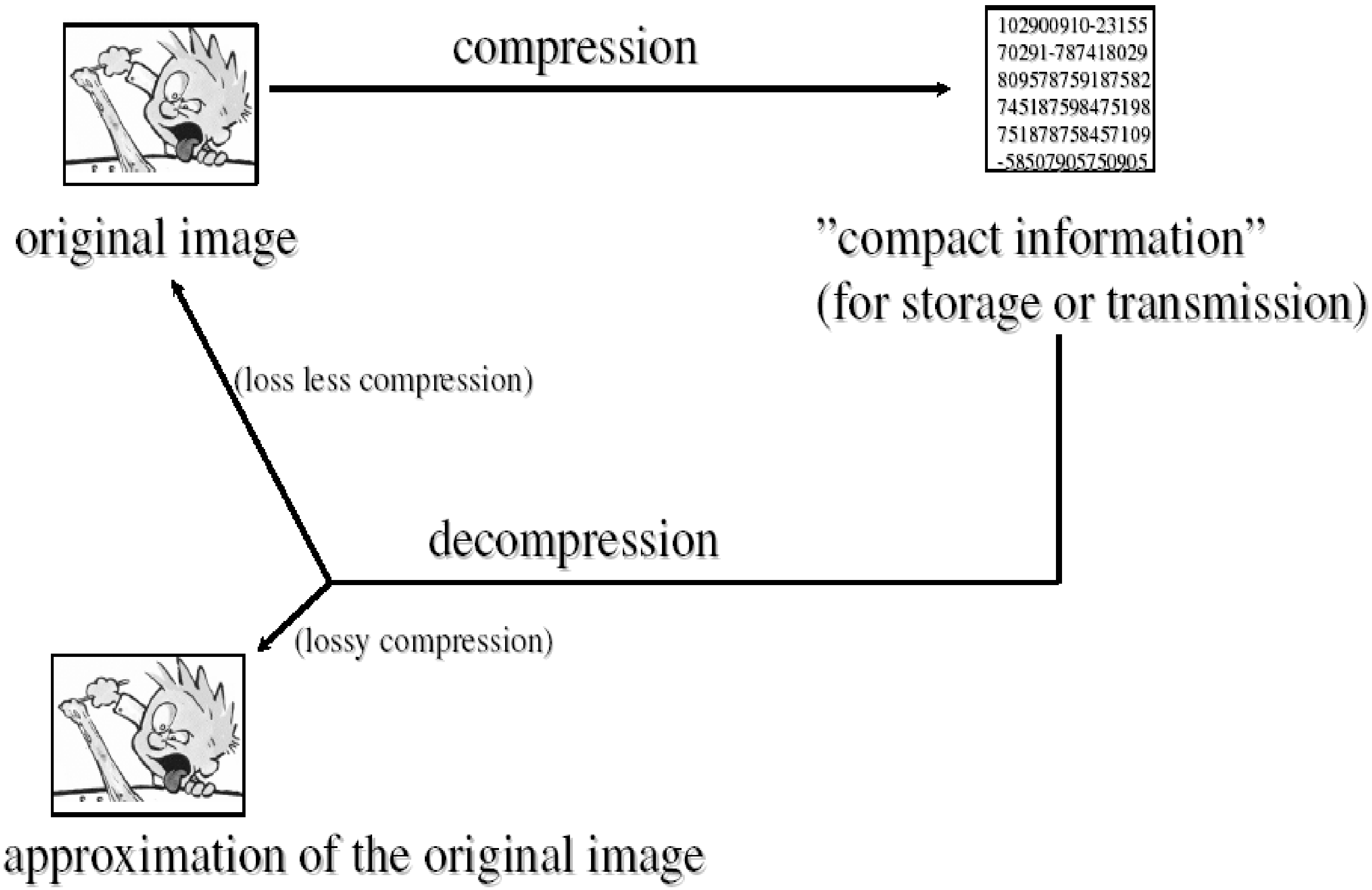
- Compress and decompress images without losing information
- Compression ratio – 2:1 to 3:1
- Especial for image archiving (storage of legal or medical records)

II. Lossy image compression

- Result in a less than perfect reproduction of the original image
- Provide higher levels of data reduction
- Compression ratio – 10:1 to 20:1or more
- Applications: –*broadcast television, videoconferencing*

# Overview

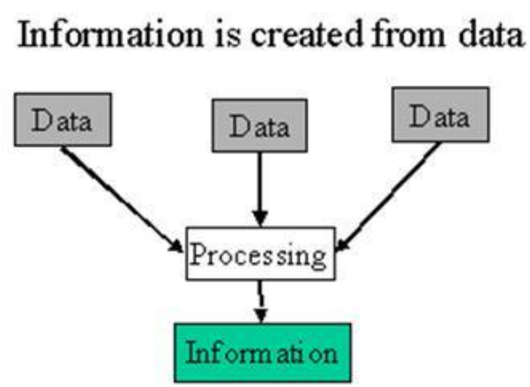



compression
102900910-23155
70291-787418029
809578759187582
745187598475198
751878758457109
-58507905750905
original image
"compact information"
(for storage or transmission)
(loss less compression)
decompression
(lossy compression)
approximation of the original image

# Data VS Information



- *Data* is not the same thing as *information*
- Data are the means to convey information; various amounts of data may be used to represent the same amount of information
- Data that provide no relevant information=*redundant data or redundancy.*
- Image compression techniques can be designed by reducing or eliminating the Data Redundancy



# Data VS Information



- Compression ratio – the ratio of the original, uncompressed image file and the compressed file

$$\text{Compression Ratio} = \frac{\text{Uncompressed File Size}}{\text{Compressed File Size}} = \frac{SIZE_U}{SIZE_C}$$

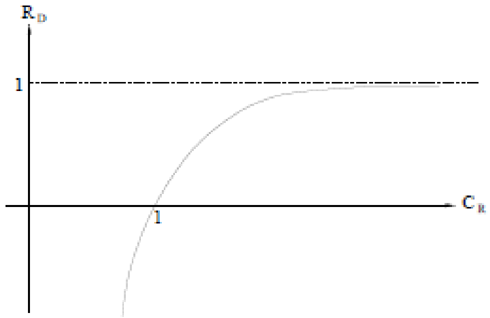
- Compression ratio = 100/10 = 10

- Compression ratio of 10 here indicates that 90% of the data in the uncompressed file is redundant

# Data Redundancy



Let $n_1$ and $n_2$ refer to amounts of data in two data sets that carry the same information




Compression ratio: $C_R = \frac{n_1}{n_2}$

Relative data redundancy: $R_D = 1 - \frac{1}{C_R} = 1 - \frac{n_2}{n_1}$
(of the first data set, $n_1$)

- if $n_1 = n_2$, $C_R = 1$ and $R_D = 0$, relative to the second data set, the first set contains no redundant data
- if $n_1 >> n_2$, $C_R \to \infty$, and $R_D \to 1$, relative to the second data set, the first set contains highly redundant data
- if $n_1 << n_2$, $C_R \to 0$, and $R_D \to -\infty$, relative to the second data set, the first set is highly compressed

$C_R = 10$ means 90% of the data in the first data set is redundant

# Data Redundancy



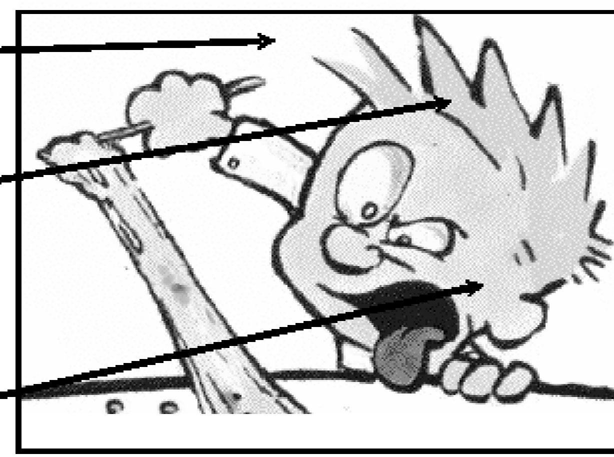
Three basic data redundancies

- Coding Redundancy
- Interpixel Redundancy
- Psychovisual Redundancy

CR: some graylevels are more common than others

IR: the same graylevel covers large areas

PVR: the eye can only resolve 32 graylevels locally



# Coding Redundancy



- A natural m-bit coding method assigns m-bit to each gray level without considering the probability that gray level occurs with: Very likely to contain coding redundancy

- **Basic concept:**

- Utilize the probability of occurrence of each gray level (histogram) to determine length of code representing that particular gray level: variable-length coding.

- Assign shorter code words to the gray levels that occur most frequently or vice versa.

# Coding Redundancy

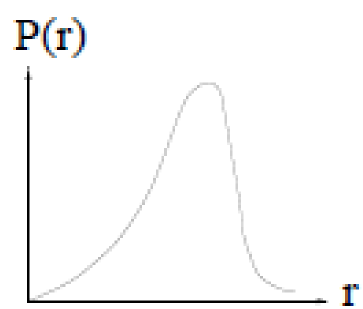


- If the gray level of an image is coded in a way that uses more code words than necessary to represent each gray level, then the resulting image is said to contain coding redundancy

- Let $P_r(r_k) = n_k / n$, (histogram) in normal 8-bit representation, each gray level $r_k$ is denoted by a code of 8-bit, L = average number of bit per pixel= 8

- If the histogram is **not** flat then it means some gray levels occur more often than others, L can be reduced by using *variable* length codes $l_k$ for gray level $r_k$

- $L = \sum_{k=0}^{L-1} P_r(r_k) \cdot l_k < 8$ if $l_k < 8$ for the more probable gray levels $r_k$



# Coding Redundancy



Let $0 \le r_k \le 1$ : gray levels (discrete random variable)

$p_r(r_k)$ : probability of occurrence of $r_k$

$n_k$ : number of pixels that $r_k$ appears in the image

$n$ : total number of pixels in an image

$L$ : number of gray levels

$l(r_k)$ : number of bits used to represent $r_k$

$L_{avg}$ : average length of code words assigned to the grey levels

$$L_{avg} = \sum_{k=0}^{L-1} l(r_k) p_r(r_k) \quad \text{where} \quad p_r(r_k) = \frac{n_k}{n}, k = 0, 1, \ldots\ldots, L-1$$

Hence, total number of bits required to code an $M \times N$ image is $MNL_{avg}$

For a natural $m$-bit coding $L_{avg} = m$.

# Coding Redundancy



Let us assume an 8-level image

8-levels: can be represented by 3-bits, m = 3.

Normalized histogram of the image is given by:

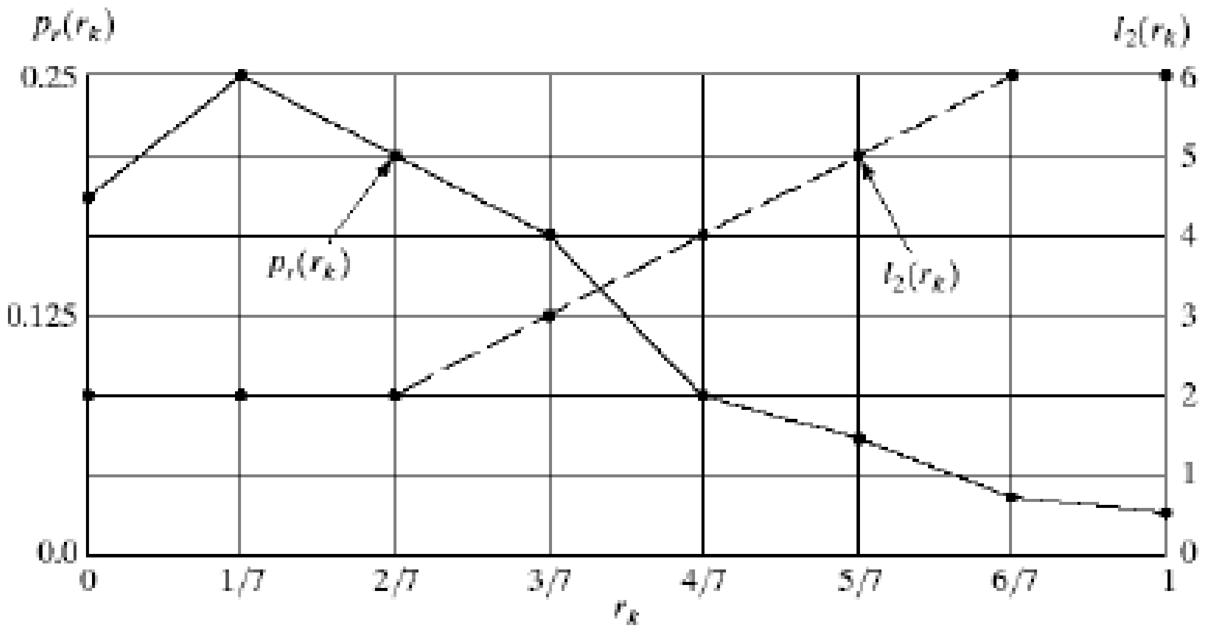



**FIGURE 8.1** Graphic representation of the fundamental basis of data compression through variable-length coding.

# Coding Redundancy



| $r_k$ | $p_r(r_k)$ | Code 1 | $l_1(r_k)$ | Code 2 | $l_2(r_k)$ |
|---|---|---|---|---|---|
| $r_0 = 0$ | 0.19 | 000 | 3 | 11 | 2 |
| $r_1 = 1/7$ | 0.25 | 001 | 3 | 01 | 2 |
| $r_2 = 2/7$ | 0.21 | 010 | 3 | 10 | 2 |
| $r_3 = 3/7$ | 0.16 | 011 | 3 | 001 | 3 |
| $r_4 = 4/7$ | 0.08 | 100 | 3 | 0001 | 4 |
| $r_5 = 5/7$ | 0.06 | 101 | 3 | 00001 | 5 |
| $r_6 = 6/7$ | 0.03 | 110 | 3 | 000001 | 6 |
| $r_7 = 1$ | 0.02 | 111 | 3 | 000000 | 6 |

**TABLE 8.1** Example of variable-length coding.

$$L_{avg_1} = m = 3 \text{ bits}, \quad \text{and}$$

$$L_{avg_2} = \sum_{k=0}^{7} l_2(r_k) p_r(r_k)$$

$$= 2(0.19) + 2(0.25) + 2(0.31) + 3(0.16)$$

$$+ 4(0.08) + 5(0.06) + 6(0.03) + 6(0.02) \ = 2.7 \text{ bits}$$

# Interpixel Redundancy



- Caused by High Interpixel Correlations within an image, i.e., gray level of any given pixel can be reasonably predicted from the value of its neighbors (information carried by individual pixels is relatively small) spatial redundancy, geometric redundancy, interframe redundancy (in general, interpixel redundancy)

- To reduce the interpixel redundancy, mapping is used. The mapping scheme can be selected according to the properties of redundancy.

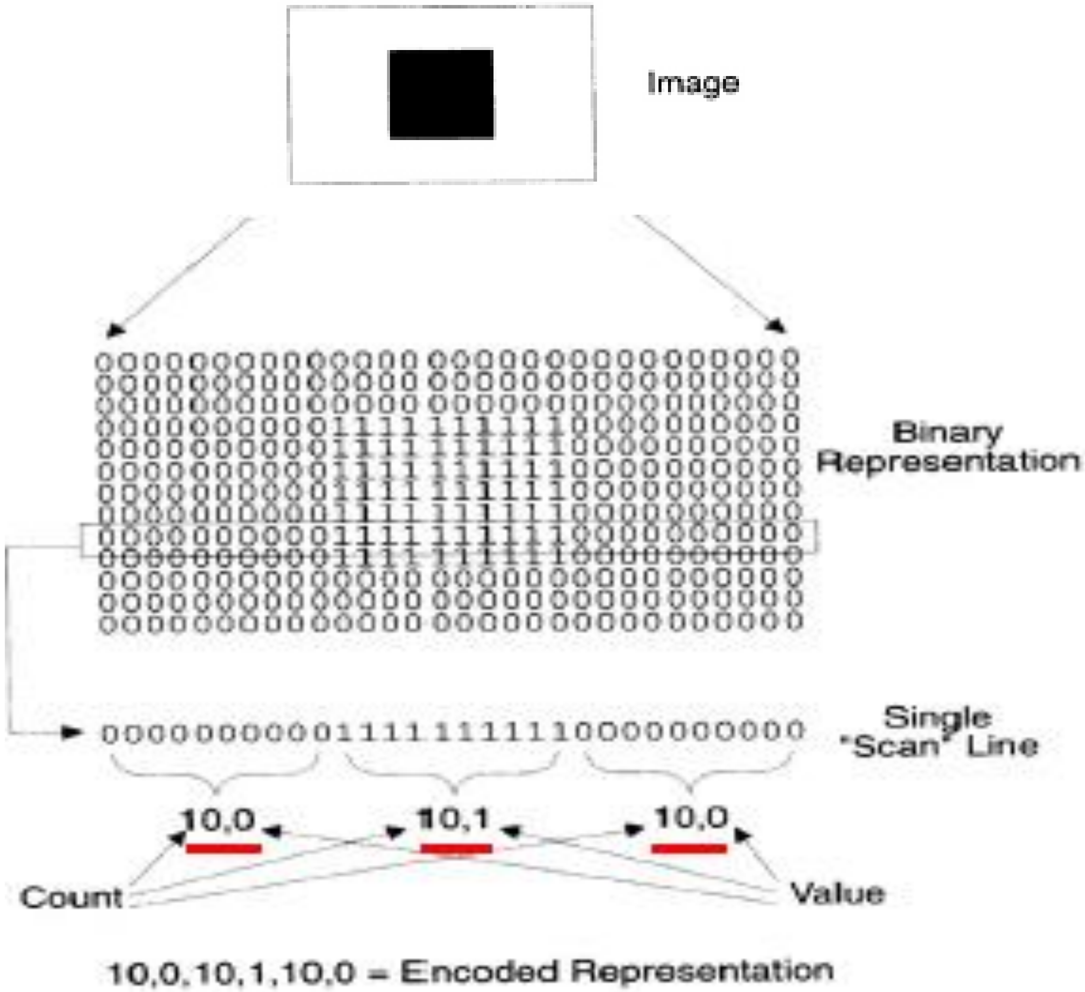
- An example of mapping can be to map pixels of an image: $f(x,y)$ to a sequence of pairs: $(g_1,r_1)$, $(g_2,r_2)$, ..., $(g_i,r_i)$, ..

  $g_i$: $i$th gray level $r_i$: run length of the $i$th run

- Removing interpixel redundancy is lossless

# Interpixel Redundancy

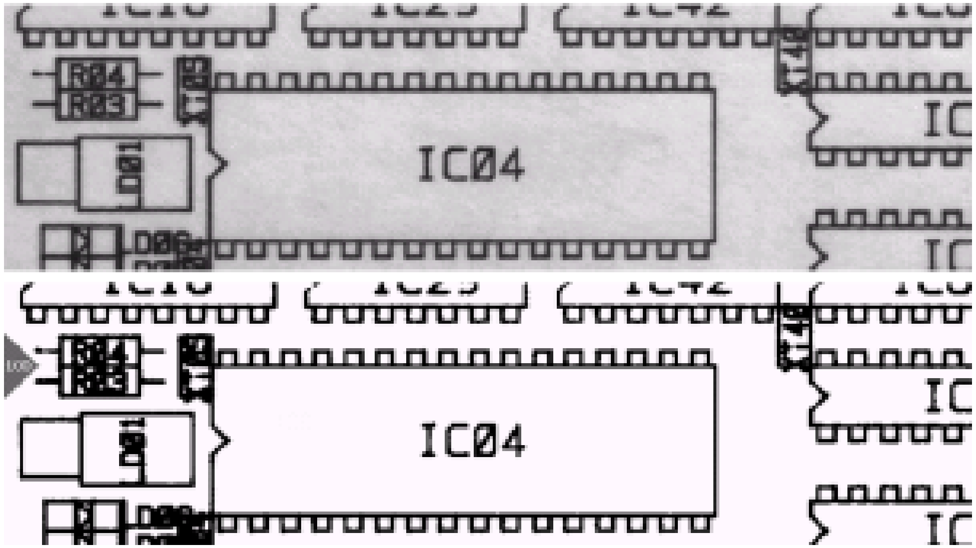




# Interpixel Redundancy



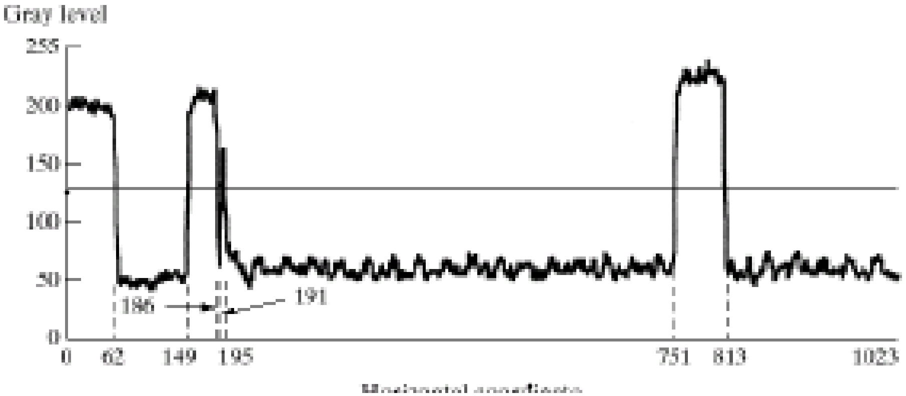
An image with size 1024($y$)×343($x$)




Consider the 1024 pixels along $x$=100 (horizontal direction)

To represent the binary image we only need 8 pairs (8 runs) because the gray level changes 7 times

Line 100:
(1,63) (0,87) (1,37) (0,5)
(1,4) (0,556) (1,62) (0,210)



# Psychovisual Redundancy

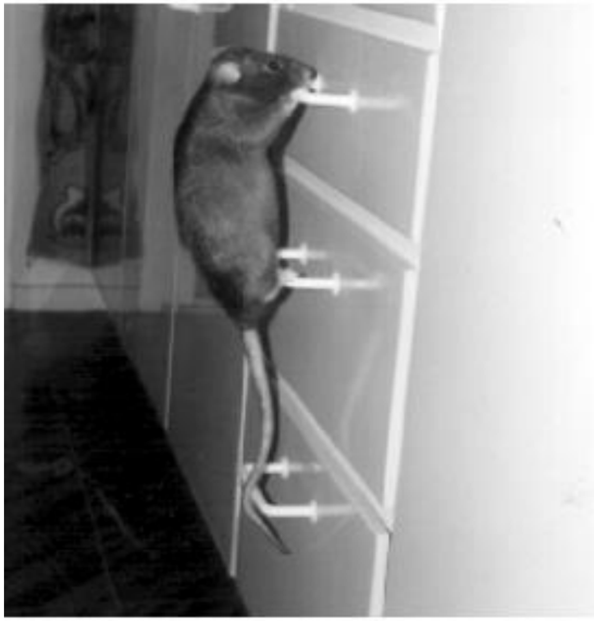


- The eye does not respond with equal sensitivity to all visual information.
- Certain information has less relative importance than other information in normal visual processing psychovisually redundant (which can be eliminated without significantly impairing the quality of image perception).
- The elimination of psychovisually redundant data results in a loss of quantitative information lossy data compression method.
- Image compression methods based on the elimination of psychovisually redundant data (usually called quantization) are usually applied to commercial broadcast TV and similar applications for human visualization.

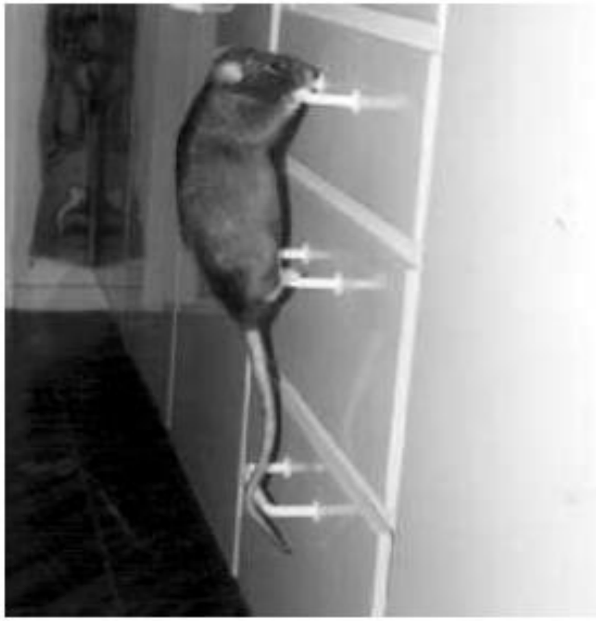
# Psychovisual Redundancy



If the image will only be used for visual observation (i.e. illustrations on the web etc), a lot of the information is usually psycho-visually redundant. It can be removed without changing the visual quality of the image. This kind of compression is usually irreversible.



0.5kB



0.05kB

# Uniform Quantizer



- Uniform scalar quantizer – the simplest quantizer

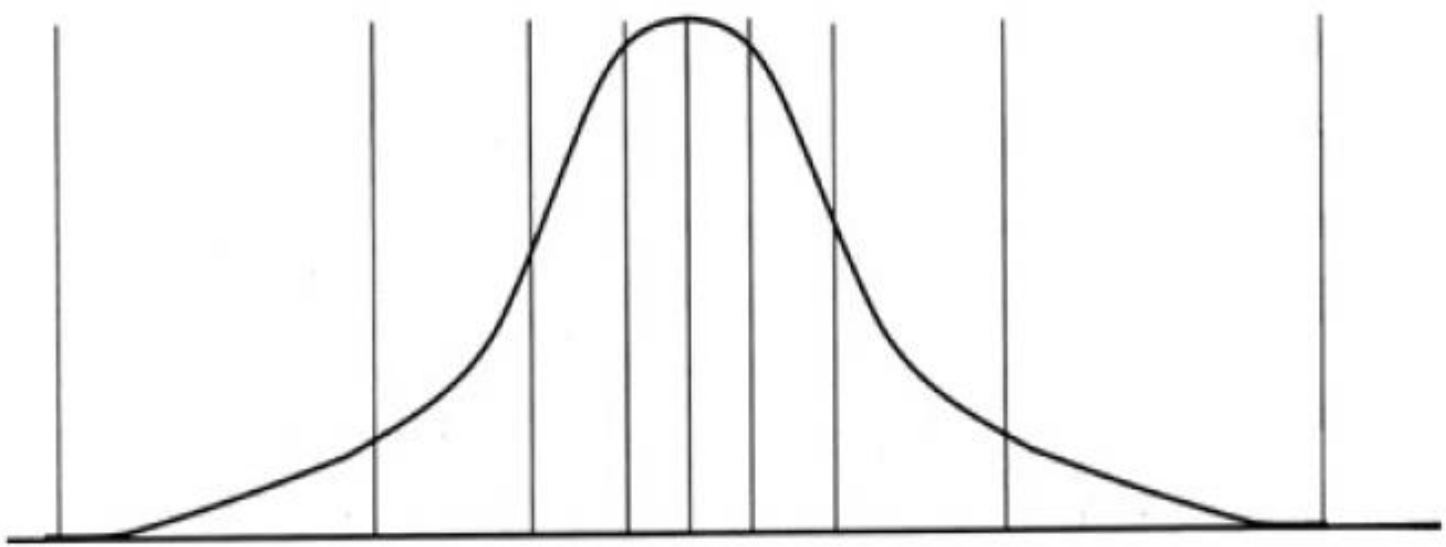
  - ✓ Masking the lower bits via an AND operation
  - ✓ Example – reduce 8-bit pixel (256 levels) to be 3-bit pixel (8 levels) by ANDing with the bit string 11100000


| Input range | Output code | Reconstruction value | MSE | Reconstruction value | MSE |
|---|---|---|---|---|---|
| 0-31 | 000 | 00000000 (0) | | 00010000 (16) | |
| 32-63 | 001 | 00100000 (32) | | 00110000 (48) | |
| 64-95 | 010 | 01000000 (64) | | 01010000 (80) | |
| 96-127 | 011 | 01100000 (96) | 325.5 | 01110000 (112) | 85.5 |
| 128-159 | 100 | 10000000 (128) | | 10010000 (144) | |
| 160-191 | 101 | 10100000 (160) | | 10110000 (176) | |
| 192-223 | 110 | 11000000 (192) | | 11010000 (208) | |
| 224-255 | 111 | 11100000 (224) | | 11110000 (240) | |

# Non-Uniform Quantizer



- Idea - the quantization levels are finely spaced in the range that gray levels occur frequently, while coarsely spaced outside of it



# Psychovisual Redundancy



Original image
(256 levels)

Normal quantization
(16 levels)

IGS quantization
(16 levels)

# Psychovisual Redundancy



- Improved Grayscale (IGS) Quantization

- If we use fewer bits/gray-levels to represent an image it causes the false contouring. To reduce this effect, a level of randomness is added to the artificial edges associated with false contouring by IGS quantization.

# Psychovisual Redundancy



- IGS quantization procedure:

    1. Set initial SUM = 0000 0000

    2. if most significant 4 bits of current pixel A = 1111

        new_SUM = A + 0000

    else

        new_SUM = A + least significant 4 bits of old SUM

| *Pixel* | *Gray Level* | *Sum* | *IGS Code* |
|---|---|---|---|
| $i$ - 1 | N/A | 0000 0000 | N/A |
| $i$ | 0110 1100 | 0110 1100 | 0110 |
| $i+1$ | 1000 1011 | 1001 0111 | 1001 |
| $i+2$ | 1000 0111 | 1000 1110 | 1000 |
| $i+3$ | 1111 0100 | 1111 0100 | 1111 |

# Fidelity Criteria



Quantify the nature and extent of information loss.

Objective fidelity criteria:

Level of information loss can be expressed as a function of the original (input) and compressed-decompressed (output) image.

Given an $M \times N$ image $f(x,y)$ (original image), its compressed-then-decompressed image: $\hat{f}(x,y)$, then the error between corresponding values are given as:

$$e(x,y) = \hat{f}(x,y) - f(x,y)$$

Total error is given by:

$$\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}\left[\hat{f}(x,y) - f(x,y)\right]$$

# Fidelity Criteria



## Subjective fidelity criteria:

-Let a number of test persons grade the images as bad/OK/good etc.

Since most decompressed images ultimately are viewed by human beings, measuring image quality by the subjective evaluations of a human observer often is more appropriate.

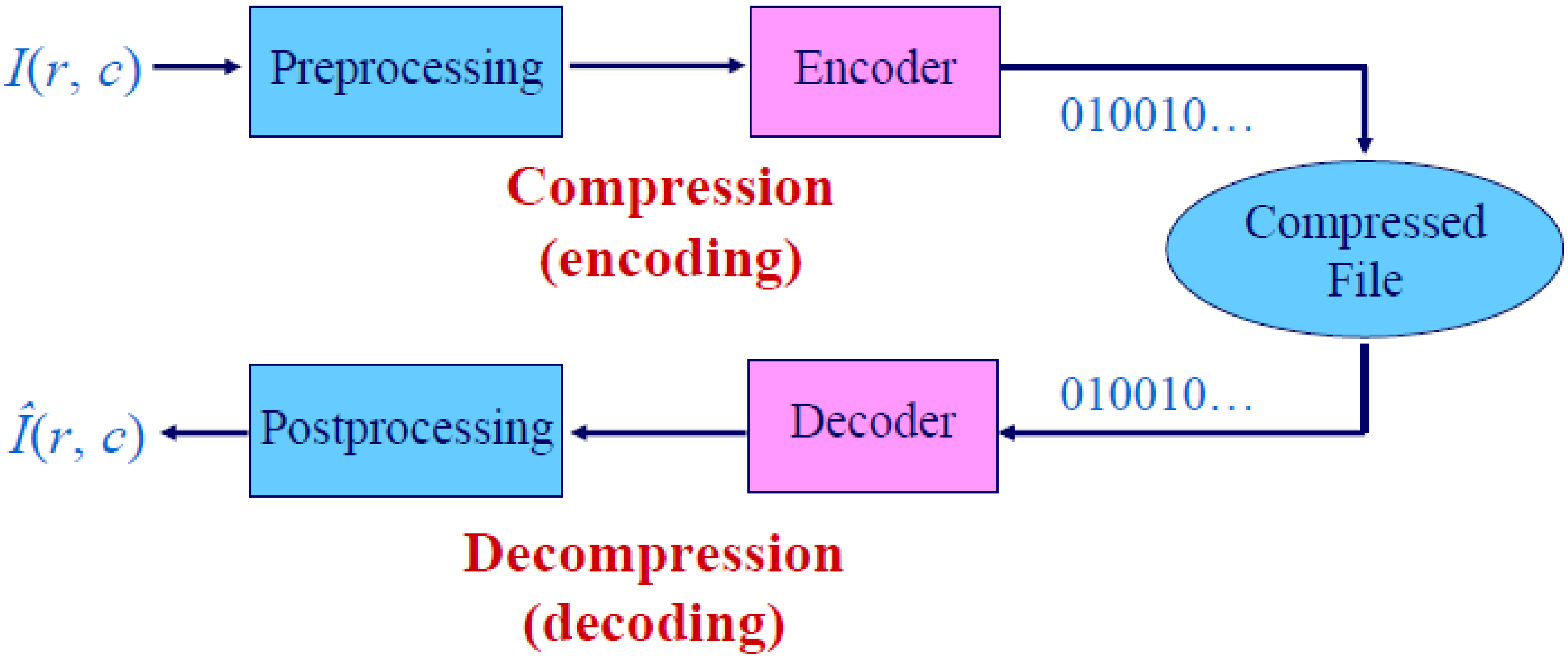
Method: evaluations are made using an absolute rating scale, by means of side-by-side comparisons of $f(x,y)$ and $\hat{f}(x,y)$.

**TABLE 8.3**
Rating scale of the Television Allocations Study Organization. (Frendendall and Behrend.)

| Value | Rating | Description |
|---|---|---|
| 1 | Excellent | An image of extremely high quality, as good as you could desire. |
| 2 | Fine | An image of high quality, providing enjoyable viewing. Interference is not objectionable. |
| 3 | Passable | An image of acceptable quality. Interference is not objectionable. |
| 4 | Marginal | An image of poor quality; you wish you could improve it. Interference is somewhat objectionable. |
| 5 | Inferior | A very poor image, but you could watch it. Objectionable interference is definitely present. |
| 6 | Unusable | An image so bad that you could not watch it. |

# Image Compression Model

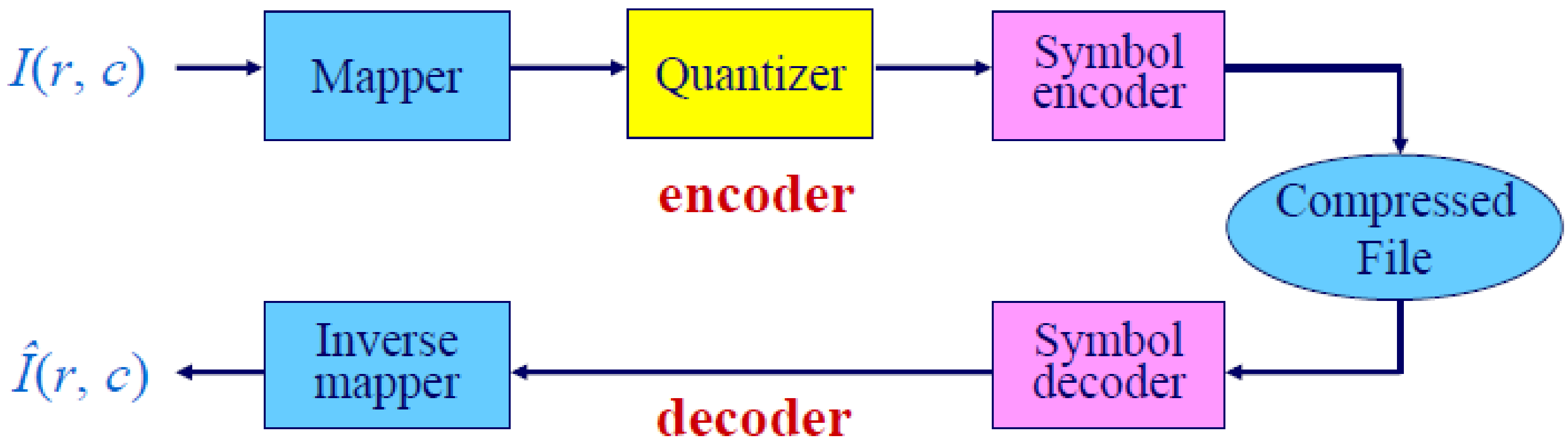





- The encoder removes input redundancies (interpixel, psychovisual, and coding redundancy)

# Image Compression Model






- mapper - reduce interpixel redundancy, reversible (run-length coding, etc.)
- quantizer - reduce psychovisual redundancy, irreversible (uniform quantization, nonuniform quantization)
- symbol encoder - reduce coding redundancy, reversible (Huffman coding, LZW etc.)

# References



1. DIP by Gonzalez

# For any query Feel Free to ask